POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Database Management System (New) | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is data abstraction in DBMS? Explain in detail. 7
   b) Consider you are asked to design a database for the Exam section of your college. Draw its ER diagram assuming required entities and their attributes. 8
2. a) Convert the ER diagram that you designed in question no 1 b) into relational schema. 7
   b) What are the views? Consider the table **tbl_emp** as follows: 8

Emp_id	Emp_name	Salary	Department	Date_of_joining
101	Anish	20000	Packing	2070-02-01
102	Rojina	18000	Cleaning	2075-04-06
103	Sita	35000	Polishing	2078-09-12

Write the SQL statements for the following:

i. Create the above table by considering Emp_id as primary key and insert the above records.
ii. Change the Department of Anish to marketing.
iii. Increase the salary of employees whose department is Cleaning by 12%.
iv. Find the name of employees having salary greater than 16000 and who joined after 2072-11-25
v. Add a new column Address to the above table.
vi. Delete the entire table.

OR

Why is joining in SQL necessary? Explain Inner Join, Natural Join and Outer Join with suitable examples.

3. a) What are the different types of integrity constraints? Explain with examples. 8

   b) What is denormalization? Why is it necessary? Explain in detail. 7
4. a) What is the role of views in security? Explain the concept of authorization with suitable examples. 7
   b) Explain the basic steps in query processing in detail. 8
5. a) Explain how the records are organized using fixed-length and variable-length records. 7
   b) What is serializability and why is it needed? Explain the ACID properties in brief. 8
6. a) What is crash recovery? Explain log-based recovery method with example. 7

OR

What is transaction rollback? Explain how the Remote Backup System provides high availability and recovery facility.

   b) List out the different categories of NoSQL databases. Explain the concept of blockchain with its properties. 8
7. Write short notes on: **(Any two)** 2×5
   a) Nested Queries
   b) Third Normal Form (3NF)
   c) Lock-based protocols for concurrency control

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Attempt all the questions.

1. a) Differentiate between schema and instances. Explain data independence with its importance. 8
   b) Design an ER diagram for an Online Food Delivery System to manage customers, restaurants, orders, and delivery. Draw the ER diagram with cardinalities and constraints. 7
2. a) What is relational algebra? Explain Selection, Projection, Union, Cartesian-product, Join and Set-Difference operations with suitable examples. 7
   b) Consider the given relational schema: 8
   Movie(Title, Director, Myear, Rating)
   Actors(Actor, Age)
   Acts(Actor, Title)
   Directors (Director, DirAge)
   Write the SQL statement for the following task:
   i. Find the movies made by Gopal Varma after 1997.
   ii. Find all actors and directors.
   iii. Find number of movie which are directed by Devid Dhaban having actor "Bishnu".
   iv. Find (Director, Actor) pairs where the directors is younger than the actor.
   v. Find the number of movie actor wise.
3. a) Explain functional dependency with its types? For R ={A, B, C, D} and F={A->BC, B->C, AB->C, AC->D}, List the closure of functional dependency F. 7

OR

What are database constraints? Explain its types with suitable examples. When do we need Denormalization? Explain.

   b) Define normalization. Create a relation in Unnormalized Form (UNF) and normalize it up to the Second Normal Form (2NF). 8
4. a) Why security is needed in database? Explain access control mechanisms. 7

OR

As a database administrator, purpose the measures to protect the database against physical and cyber threats.

   b) Define ACID properties. Explain view serializability with suitable examples. 8
5. a) What do you mean by query processing? Explain the steps of query processing with necessary diagram. 7
   b) Construct a B+ tree for the following set of key values: {2,5,7,9,11,17,21,23,29,31}. Assume that the tree is initially empty and values are added in ascending order where the pointer number is FOUR. 8
6. a) Consider following two transactions T1 and T2 given in figure where T1 executes before T2. Also consider initial value of A, B and C are 1000, 2000 and 3000 respectively. 7

T1	T2
Read (A)	Read (C)
A = A - 500	C = C – 200
Write (A)	Write (C)
Read (B)	
B = B + 500	
Write (B)	

   Use the concept of deferred and immediate database modification log based recovery method in given problem.

   b) Define data warehousing and explain how it supports decision-making processes in large organizations. 8
7. Write short notes on: **(Any two)** 2×5
   a) Query cost estimation
   b) Set operations in SQL
   c) NoSQL Databases and its characteristics

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
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Course: Database Management System (New) | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) What is a database management system? Difference between Physical and Logical Data Independence. 7

   b) Draw E-R diagram for airline reservation system. The system must keep track of customers and their reservations, flights and their status, seat assignments on individual flights. Include appropriate relationship and cadinality. 8

   OR

   Draw an ER diagram of an "Exam System of Pokhara University" using extended features. Explain Primary key and Foreign key.

2. a) Consider the following schema: 7

   STUDENT(Student_ID, Student_name, Major)
   COURSE(Course_ID, Course_name, Credits)
   ENROLLS_IN(Student_ID, Course_ID, Grade)

   Write the relational algebra expressions for the following cases:

   i. Find the names of all students who are enrolled in courses with names that start with "C".
   ii. Update the grades of students to 'A' whose current grade is 'B'.
   iii. Find the average credits of courses each student is enrolled in.
   iv. Update the name of the course "Physics" to "Advanced Physics".

   b) Let us consider the following relation: 8

   Sailors (sid, sname, rating, age)
   Boats (bid, bname, color)
   Reserves (sid,bid, day)

   Write a SQL statements for the following:

   i. Find the records of sailors who have reserved boat number 75 (bid=75).

   ii. Update the color of the yellow boat to green.
   iii. Delete the records of sailors whose rating is less than 5.
   iv. Find the name and rating of sailors who have reserved a black or blue boat.
   v. Find the age of sailors who have reserved boat number 25 on day 3.

3. a) Define integrity constraints? Explain different integrity constraints with suitable examples. 8

   OR

   Why is database normalized? Explain about 1NF, 2NF, 3NF & BCNF.

   b) Explain different types of functional dependencies with examples. Find the closure of attributes A, AC and ABC for the relation R(A,B,C,D) where $A \rightarrow B, BC \rightarrow D$ are the dependencies in R. 7

4. a) Compare authentication and authorization. Explain the different types of access control mechanisms. 7

   b) How query optimization is carried out? Explain about cost estimation of query. 8

5. a) Construct a B+ tree of order 4 for the given key values assuming empty tree initially and keys arranged in ascending order: {4,8,1,20,5,15,16,7,9,38,34,6,39,25} 7

   b) What is serializability? Explain how concurrency control protocols are used? 8

6. a) What are NoSQL databases? Explain the advantages of NoSQL databases. 7

   b) Differentiate between centralized and distributed databases. Explain in brief about NOSQL. 8

7. Write short notes on: (Any two) 2×5

   a) Equivalence of expressions
   b) Stored Procedures
   c) ACID Property

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Attempt all the questions.

1. a) Describe the architecture of a Database Management System and Database Application (1-tier vs 2-tier vs 3-tier). 7

   b) Create an ER diagram for a movie rental store where: 8

      i. Customers rent movies.

      ii. Movies have movie ID, title, genre, and release year.

      iii. Each customer has a customer ID, name, and membership type.

      iv. Capture the rental date and return date for each movie rented.

2. a) Consider the following relational database where the primary keys are underlined: 8

   Empdetail (<u>emp-name</u>, street, city)

   Jobdetail (<u>emp-name</u>, university-name, salary)

   Universitydetail (<u>university-name</u>, city)

   Headdetail (<u>emp-name</u>, head-name)

   Express the following queries in relational alzebra:

      i. Give all employees of Pokhara University a 15 % salary raise.

      ii. Update the city of Amit to Butwal.

      iii. Give all managers in this database a 25 percent salary raise.

      iv. Find all employee name who work for Pokhara University and has salary greater than 70000.

   b) Consider the relational database with the following relations: 7

   Student(StudentID, Name, DeptID, Age)

   Department(DeptID, DeptName, HOD)

   Enrollment(EnrollID, StudentID, CourseName, Grade)

   Now write the SQL command for the following:

      i. Write SQL statements to create the student table.

      ii. Display each student's name, department name, and course name for all enrolled students.

      iii. Display the average grade for each course, based on the Enrollment table and group the result by CourseName.

      iv. Update the age of student 'Iris' to 26.

3. a) What is Domain, Entity Integrity, and Referential Integrity constraints? Explain with examples. 8

   OR

   Define normalization and explain its purpose in database design. Discuss the problems (anomalies) that occur in unnormalized databases.

   b) What is relational model? What are the features of good relational database design? Explain. 7

4. a) Explain access control, authorization, and views in database security. 7

   b) What is an execution plan? Describe how equivalence rules are used in optimizing queries in DBMS. 8

5. a) Define indexing. Differentiate between B+ Tree Indexing and Hash Indexing. 7

   OR

   Discuss the organization of records in Heap and Indexed Sequential file systems.

   b) What is Serializability? Explain the lock-based concurrency control. 8

6. a) What is a transaction log? How are undo and redo operations handled using log records? Explain. 8

   b) Discuss the advantages and limitations of NoSQL databases. Explain different categories of NoSQL. 7

7. Write short notes on: **(Any two)** 2×5

   a) Data Independence

   b) DDL and DML

   c) Denormalization

POKHARA UNIVERSITY

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Programme: BE | Full Marks : 100
Course: Database Management System (New) | Pass Marks : 45
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Attempt all the questions.

1. a) Explain the three levels of data abstraction in DBMS with suitable examples. Also define data independence and why it is important. 7
   b) Design an ER Diagram for a "Hospital Management System." Your diagram must include: 8
      i. At least four entity sets (e.g., Patient, Doctor, Room, Medicine).
      ii. Examples of multi-valued and derived attributes.
      iii. Clearly marked participation constraints and cardinalities.
2. a) Given the relations: 7
   Student(SID, Name, Major)
   Course (CID, CName, Credits)
   Enroll(SID, CID, Grade)
   Write relational algebra expressions for the following:
      i. Find the names of students who have a grade of 'A'.
      ii. List all SIDs of students who have not enrolled in any course.
      iii. Perform a Natural Join between the two tables.
      iv. Find the names of students enrolled in 'CS101'.
   b) Consider the relational database 8
   Employees (EmpID, Name, DeptID, Salary, JoinDate)
   Departments (DeptID, DeptName, Location)
   Projects (ProjectID, ProjectName, Budget, DeptID)
   Write SQL queries for:
      i. Create the table Employees with appropriate constraints.
      ii. Find the Name, Salary of employee of 'IT' department.

      iii. Increase the salary of employees in the "IT" department by 10%.
      iv. Delete all projects with a budget less than 40,000.
3. a) Explain about different types of constraints in Database? Explain each with suitable example. 7
   b) Explain about different anomalies we face before normalization? Also explain about 2NF, 3NF, BCNF with suitable example. 8

   OR

   What is closure of attribute sets? How is it computed? For $R = (A, B, C, D, E, F)$ and $F = \{A \rightarrow BC, E \rightarrow CF, B \rightarrow E, CD \rightarrow EF\}$, Compute $(AB)^+$.
4. a) What is access control in a database system? Explain different access control mechanisms used in DBMS. 7
   b) What is Query Processing? Explain the steps involved in query processing with suitable diagram. 8
5. a) Describe different file organization techniques used in DBMS. Explain heap file organization with its merits and demerits. 7
   b) What is Conflict Serializability? Explain how the Two-Phase Locking (2PL) protocol ensures that a schedule is serializable. 8

   OR

   Explain Lock Based Protocol along with its pitfalls. Let we have two accounts A and B with values 5000 and 10000 respectively. Transaction T1 transfers 1000 from A to B. Transaction T2 display sum of A and B. Apply appropriate Locks in above transaction T1 and T2 and make serial schedule.
6. a) Describe the Log-Based Recovery mechanism. Differentiate between Deferred Database Modification and Immediate Database Modification techniques. 7
   b) What is Object Relational Mapping (ORM)? Explain its role in modern application development. 8
7. Write short notes on: **(Any two)** 2×5
   a) NoSQL Databases and its characteristics
   b) Transaction State Diagram
   c) Stored Procedure

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2019
Programme: BE Full Marks: 100
Course: Database Management System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) What do you understand by Data Independence? How is Schema different from Instance? Justify with some suitable examples. 7

   b) How does UML diagram assist during data modeling? Draw an E-R diagram for a Gandaki Auto Vehicle Shop System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram 8

2. a) How Relational Algebra is different from Relational Calculus? Define TRC and DRC. 7

   b) Consider a simple relational database of Hospital Management System. (*Underlined attributes represent Primary key attributes*) 8

   Doctors (<u>DoctorID</u>, DoctorName, Department, Address, Salary)
   Patients (<u>PatientID</u>, Patent Name, Address, Age, Gender)
   Hospitals (<u>PatientID</u>, <u>Doctor ID</u>, HostpitalName, Location)

   Write down the SQL statement for the following:

   i. Display ID of Patient admitted to hospital at Pokhara and whose name ends with's'.
   ii. Delete the record of Doctors whose salary is greater than average salary of doctors.
   iii. Increase the salary of doctors by 18.5% who works in OPD department.
   iv. Find the average salary of Doctors for each address who have average salary more than 55K.

3. a) Define Normalization. Explain about 1NF, 2NF & 3NF. 7

   b) What do you mean by decomposition of relational schema? Suppose we are given Schema R = {A,B,C,G,H,I} and set of functional dependencies F={A→ B,A→ C, CG→ H, B→H, CG→I}. Find the closures of functional dependency F. 8

4. a) What is Access control mechanism in database? Explain different types of access control mechanism. 8

   b) Diagrammatically illustrate and discuss the steps involved in processing a query. 7

5. a) Construct a B+ tree for the following set of key values: (2,3,5,7,11,17,19,23,29,31) Assume that the tree is initially empty and values are added in ascending order where the pointer number is Four 8

   b) What is Crash Recovery? What are the problems due to crash? How the problems can be avoided, explain any one briefly. 7

6. a) When does deadlock occurs? Explain two-phase commit protocol with example. 7

   b) What are data fragmentations? State the various fragmentations with examples. 8

7. Write short notes on: **(Any two)** 2×5

   a) ACID property
   b) QBE
   c) Object Relational Model

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2019
Programme: BE Full Marks: 100
Course: Database Management System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
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Attempt all the questions.

1. a) Explain the concept of DBMS and its applications tracing the evolution. 7
   b) Construct an ER diagram for keeping records for Library Management Systems. 8
2. a) Using the following schema represent the following queries using Relational algebra : 8
   PROJECT (Project num, ProjectName, ProjectType, ProjectManager)
   EMPLOYEE (Empnum, Empname)
   ASSIGNED_TO (Projectnum, Empnum)
   i) Find Employee details working on a project name starts with 'L'
   ii) List all the employee details who are working under project manager "Rohan"
   iii) List the employees who are still not assigned with any project.
   iv) List the employees who are working in more than one project.
   b) Write the SQL statements for the following queries by reference of **Hotel_details** relation: 7

hotel_id	hotel_name	estb_year	hotel_star	hotel_worth
1	Hyatt	2047	Five	15M
2	Hotel Ktm	2043	Three	5M
3	Fulbari	2058	Five	20M
4	Yak and Yeti	2052	Four	11M
5	Hotel Chitwan	2055	Three	7M

   i. Create a database named hotel & table relation.
   ii. Create a view named Price which shows hotel name & its worth.
   iii. Modify the data so that Hotel Chitwan is now four star level.
   iv. Delete the records of all hotels having worth more than 9M.

3. a) What are store procedures? Explain equi Join, natural join, left and right outer join with examples. 8
   b) Differentiate between Functional Dependency and Multi Valued Dependency? Explain closure set of functional dependencies with example. 7
4. a) Define third normal form. Convert the following 2NF relation into 3NF(consider **Name** as primary key) 8

Name	Address	Phone	Salary	Post
Gill	KTM	456789	20000	Engineer
Van	BKT	654321	20000	Engineer
Robert	KTM	456789	20000	Engineer
Brown	BKT	654321	10000	Overseer
Albert	KTM	454545	10000	Officer

   b) What is security and integrity violations? Explain the need of access control, Authorization and Authentication. 7
5. a) What is query cost estimation? Explain cost based & heuristic based choice of evaluation plan for query optimization. 7
   b) Create a B+ tree of order 4 with following data: 8
   (4, 9, 16, 25, 1, 20, 13, 15, 10, 11, 12) of order 4. Assume that, tree is initially empty and values are added in ascending order.
   Also, show the formation of tree after the deletion of 16.
6. a) What is concurrency control? Describe ACID property of transaction. 8
   b) Define recovery. When the two transactions are said to be in deadlock state? How these deadlocks can be addressed. 7
7. Write short notes on: (Any two) 2×5
   a) Architecture of Distributed Database
   b) Role of Database administrator
   c) Dense and Sparse Index

Com - 4th

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2020
Programme: BE	Full Marks: 100
Course: Database Management System	Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why data independence is important in data modeling? Differentiate between physical and logical data independence. 7

   b) Define and explain benefits of data model. Draw an E-R diagram for a Vehicle Management System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 8

2. a) Explain Relational Algebra. What are the relational algebra operations that can be performed? Give an example of all. 7

   b) Write SQL statements for following: 8

   i. Create a table named Automotor with chasis_number as primary key and following attributes: veh_brand, veh_name, veh_model, veh_year, veh_cost, veh_color, veh_weight

   ii. Enter a full detailed information of an automotor.

   iii. Change any Automotor's year to 2019.

   iv. Remove all Automotor records whose model contains character 'i' in last position.

   v. Display the total cost of all vehicles of the table Automotor.

   vi. Create a view from above table having vehicles only red color.

   vii. Display details of Automotor ordering on descending manner by brand name and by ascending on model when brand matches.

   viii. Change data type of color so that it only takes one character.

3. a) Differentiate between join and sub query. Explain different SQL joins with examples. 8

   b) What is functional dependency? Discuss its types. Explain the role of Functional dependency in the process of normalization. 7

4. a) What is multi-valued dependency? Illustrate the advantage of 4NF with suitable example. 8

   b) Describe the GRANT functions and explain how it relates to security. What types of privileges may be granted? How rights could be revoked? 7

5. a) Define query optimization. What are the basic steps of query processing? Explain. 7

   b) In terms of file organization, define *Indexing, Elevator Algorithm, Log disk.* How does a mechanical hard disk work? 8

6. a) What is a transaction? What is a serializable schedule? Describe the dead lock handling mechanism. 7

   b) Explain different types of crash recovery algorithm with suitable examples. 8

7. Write short notes on any two: 2×5

   a) Two phase locking

   b) Data Godown v/s Data Warehouse

   c) Schema and instances

POKHARA UNIVERSITY

Level: Bachelor | Semester – Spring | Year: 2020

Program: BE | Full Marks: 70

Course: Database Management System | Pass Marks: 31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all questions

Section - A: (5×10=50)

Q. N. 1 Illustrate redundancy and the problems that it can cause? Compare and Contrast file Systems with database systems? 10

Q. N. 2 Explain the importance of data modeling. Draw E-R diagram for a departmental store that keeps the information about customer, supplier and item. 10

Q. N. 3 List all security issues you are familiar that can breaches security. Encryption and Decryption are very important part in cryptography so justify this statement with suitable example. 10

OR

What is the importance of indexing in file organization? Write in detail about Hash based Indexing and Tree based Indexing? 10

Q. N. 4 What are equivalent query explain with example. Explain the steps in query processing with diagram in detail.

Q. N. 5 Explain ACID properties of transactions with examples of each property. Also explain the importance of concurrency control in transaction processing. 10

Section - B: (1×20=20)

Q. N. 6 Write both relational algebra and SQL statement for the following schemas (underline indicates primary key): 20

Employee (Emp_No , Name , Address)
Project (PNo ,Pname)
Workon (Emp_No , PNo)
Part (Partno ,Part_name , Qty_on_hand)
Use (Emp_No , PNo , Partno ,Number)

a. Listing all the employee details who are not working yet .
b. Listing partname and Quantity on hand those were used in DBMS project.
c. List the name of the projects that are used by employee from London.
d. Modify the database so that Jones now lives in USA.
e. Update address of an employee 'Japan' to 'USA'.

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021

Programme: BE | Full Marks: 100

Course: Database Management System | Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is data independency? Why is it required in DBMS? Explain in detail. 7

   b) What is ER diagram? Draw an ER diagram for a library system. Assume the entities- student, teacher, book and semester. In the diagram illustrate the concept of strong entity, week entity, composite attributes, multivalued attributes, and derived attributes. 8

2. a) Define schema and views. Considering the following schemas: 8

   Sailors (sid, sname, rating, age)
   Boats (bid, bname, color)
   Reserves (sid, bid, day)

   Write relational algebra expressions for the following queries:
   i. Find the records of sailors who have reserved boat number 103 (bid=103).
   ii. Update the color of the boat, where bid is 104, into green.
   iii. Find the names of sailors who have reserved a red or green boat.
   iv. Find the names of sailors who have reserved boat number 103 on day 5.
   v. Find the names of sailors whose name is not 'Ram'
   vi. Find the names of all boats.

   b) What are DDL and DML queries in SQL? Consider the relations in 2(a) and write the SQL statements for the queries in 2(a). 7

3. a) What are the different types of integrity Constraints? Explain with examples. 8

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2022
Programme: BE Full Marks: 100
Course: Database Management Systems Pass Marks: 45
Time : 3hrs.

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Attempt all the questions.

1. a) Define DBMS. What are the advantages of DBMS over traditional file system? 7
   b) Why do you need E-R diagram? Draw an E-R diagram for online shop management system. Assume relevant entities and attributes for the given system. 8
2. a) Suppose we have the following relation: 8
   Employee(person_name, street, city)
   Works (person_name, company_name, salary)
   Company (company_name, city)
   Write relational algebra expressions for the following queries:
   i. Find names of all employees who live in 'Butwal' and whose salary is less than Rs. 50,000.
   ii. Find the names of all employees who work for "Nabil Bank Limited".
   iii. Find the names and cities of residence of all employees who work for "Global bank"
   iv. Update the salary of all employees by 10%.
   b) Define stored procedure. What is the advantage of the stored procedure? Explain how stored procedures are created in SQL. 7
3. a) Consider the relation **Actress_Details** and write the SQL statements for the following queries: 8

Players_id	Actress_name	Debut_year	Recent_release	Actress_fee
1	Renu	2010	Samay	400000
2	Sita	2022	Radha	300000
3	Geeta	2001	Mato	600000
4	Amita	1990	Man	700000
5	Karishma	1989	Prem	100000

   b) What is database normalization? Discuss normalization process with a suitable example until it satisfies 3 NF. 7
4. a) What are the needs of security? Explain about the access control, authorization and authentication. 7
   b) Consider the relation schema in 2(a). Write the relational algebra expression for the query "Find the names of sailors who have reserved a red or green boat". Construct the initial operator tree and final efficient operator tree after applying transformation rules. 8
5. a) Explain file organization using hash indices with example. 7
   b) What is Crash Recovery? Explain log based recovery method with example. 8
6. a) Explain the serial schedule and serializable schedule with examples. 8
   b) What are object- oriented database model? Explain the advantage and disadvantage of object-oriented database over relational database. 7
7. Write short notes on: **(Any two)** 2×5
   a) Data dictionary
   b) ACID properties
   c) Query By Example (QBE)

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BCA | Full Marks: 100
Course: Database Management System | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is DMBS? Compare DMBS with conventional file system. 7

   b) Design an ER diagram for an **Online Clothing Shopping app** assuming necessary entities and attributes. Explain the entities and their relationships in the diagram. 8

2. a) Consider the following relations: 8
   - employee (eid, name, address, salary)
   - works (eid, cid)
   - company (cid, name, address)

   Write Relational Algebraic Statements for below queries:
   i. Find the employees who are earning less than 20,000.
   ii. Display employees working for **'ABC Soft Pvt Ltd'** company.
   iii. Delete company records from **California**.
   iv. Upgrade employee's salary by **1,000** who are from **Pokhara**.

   b) Consider the following relations: 7
   - Book (id, title, author, publisher, year, price)
   - bookstore (name, address, city)
   - sold (name, id, quantity)

   Write the SQL statements for the following queries.
   i. Give the details of book whose author name is 'Prativa' and book title is 'Iris'.
   ii. Display book title and price whose sold quantity is more than 500.
   iii. Insert a new book details.

3. a) What do you mean by join? Explain inner join and outer join with suitable example.? 7

8

i. Create the table Actress_details relation.
ii. Delete the data of actress whose recent release is Prem.
iii. Modify the database so that Renu's new release is "Win the Race" film
iv. Insert a new record in the above table.

b) Consider the following relation where {M_ID and P_ID} are primary keys. State in which normal form the relation is. What anomalies can occur in this relation? How can these anomalies be removed? 7

M_ID	M_Date	P_ID	Quantity
M11	16 June, 2022	11	20
M11	26 June, 2022	16	30
M22	3 September, 2022	15	20
M22	13 September, 2022	16	60
M22	23 September, 2022	12	35

4. a) When do you use triggers? Explain with any one example of triggers in SQL. 7

   b) Differentiate between authentication and authorization. How encryption and decryption occurs in private key and public key cryptography? 8

5. a) Consider the relation schema in 2(a). Write the relational algebra expression for the query "Find the names of all employees who lives in Pokhara". Construct the initial operator tree and final efficient operator tree after applying transformation rules. 8

   b) What is file organization? Explain how you organize files using hash index. 7

6. a) What is crash recovery? Discuss shadow paging with necessary diagram. 7

   b) What do you understand by concurrency control? Explain two phase locking protocol with examples. 8

7. Write short notes on: **(Any two)** 2×5
   a) Remote backup system
   b) ACID properties
   c) Distributed database

b) Why do we need data normalization? Explain 1NF and 2NF and 3NF with examples.

4. a) Define database constraints. Explain integrity constraint and referential constraint. 8

b) What are the importances of security in DBMS? Explain encryption and decryption techniques. 7

5. a) What is Query Processing? Explain query optimization and query cost estimation in detail. 8

b) What is the benefit of indexing? Explain index sequential file organization in detail. 7

6. a) What is crash recovery system? Explain shadow paging recovery technique. 7

b) Why do you need concurrency control? Explain how 2PL protocol used in transaction. 8

7. Write short notes on: (Any two) 2×5

a) DDL and DML in SQL

b) Triggers

c) Hash Index

POKHARA UNIVERSITY

Level: Bachelor Semester : Spring Year : 2023
Programme: BE Full Marks: 100
Course: Database Management System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Differentiate between database schema and instances. Briefly describe DDL, DML and DCL. 8

b) Define relational data model. Draw an E-R diagram for a Library Management System including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 7

a) Suppose we have the following relation. 8

Employee(person_name, street, city)
Works (person_name, company_name, salary)
Company (company_name, city)

Write relational algebraic expressions for the following queries:

i. List the name and city of employee who work in "pokhara" and have salary greater than Rs. 50,000.
ii. Find the names of all employees who work for "ABC bank".
iii. Delete all employee who come from "Chitwan".
iv. Increase salary of all employee by 15%.

b) What are different kinds of joins? Explain in brief. 7

a) Write SQL statements for the following queries using the given Employees relation: 8

E_id	Fname	Lname	Department	Salary	Hire_Date
01	Ramu	Bashyal	Sales	20000	2023-08-08
02	Damu	Pandey	IT	50000	2022-01-01
03	Biru	B.K.	Sales	40000	2021-02-10
04	Hiru	Dhamala	HR	35000	2023-12-18
05	Biren	Khadka	IT	60000	2012-10-22

i. Create a database named Company and Employees relation.
ii. Create a view that shows the E_id, Department and Hire_Date of all employees.
iii. Modify the table such that the Department of Biren is HR now.
iv. Delete the record of employees whose Lname is "Pandey".

b) What is referential integrity? Explain about a trigger with an example. 7

4. a) What is database normalization? Explain in detail about 1NF, 2NF, 3NF with suitable examples. 8

b) What are authorization and authentication? Why are they important? Explain in detail. 7

5. a) What are the steps in query processing? Make an operator tree for the following SQL expression: 7

Select customer_name
FROM branch, account, depositor
WHERE branch_city='btl' AND balance>2000;

b) What are the benefits of using B Tree index over the sequential and indexed sequential file organization? Explain. 8

6. a) Explain Log based recovery system with an appropriate log record example. 7

b) Why should the transactions' schedule be serialized? Explain conflict and view serializability with example. 8

7. Write short notes on: (Any two) 2×5

a) Data Dictionary
b) Stored procedure
c) Object oriented Database

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2024
Programme: BE Full Marks: 100
Course: Database Management System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define schemas and instances. What are the major functions of Database Manager? Discuss. 7

b) Explain the benefits of data model. Construct an ER diagram for keeping records for Library Management System. 8

2. a) Consider the following schema: 8

EMPLOYEE (Employee_ID, Employee_Name, Department)
PROJECT(Project_ID,Project_Title, Budget)
WORKS_ON(Employee_ID, Project_ID, Hours_Worked)

Write the relational algebra expression to:

i. Find the names of all employees who are working on a project with a project title that contains the word 'Development'.
ii. Update the budget of projects to 1,000,000 where the budget is currently less than 500,000.
iii. Find the total hours worked by each employee across all projects.
iv. Update the name of the 'HR' department to 'Human_Resources'.

b) Write the SQL statements for the following queries by reference of Hotal_details relation. 7

3. a) What are advantages of normalization? Illustrate importance of normalization with suitable examples. 8

b) Why Integrity Constraints are used? Explain types of integrity constraints with examples. 7

4. a) Differentiate between authorization and authentication? Explain about private key cryptography with example. 8

b) What is query processing? Explain its steps with block diagram. 7

5. a) Create a B+ tree of order 4 with the following data: (1,4,7,10,17,21,31,25,19,20,28) 7
   b) Explain log based recovery and different methods of keeping log along with example. What is shadow paging? 8
6. a) What is view serialize ability? Explain 2PL and time stamping. 7
   b) What are distributed database systems? Explain advantages and disadvantages of object oriented database model. 8
7. Write short notes on: (Any two) 2×5
   a) Data Abstraction
   b) ACID
   c) Triggers

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2024
Programme: BE		Full Marks : 100
Course: Database Management System		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Data Independency. Differentiate between Schema and Instance with the help of an example. 7
   b) Draw an E-R diagram for an online shopping system. Assume necessary entities and attributes. Also, the system should model the following: 8
      - A customer can place many orders, and each order contains many items.
      - An order can be shipped to different addresses, and each shipping address can be linked to multiple orders.
      - The system tracks payment status and the payment method for each order.
2. a) Using the following schema represent the following queries using Relational algebra: 8

   Doctors (DoctorID, DoctorName, Department, Address, Salary)
   Patients (PatientID, Patent Name, Address, Age, Gender)
   Hospitals (PatientID, Doctor ID, HostpitalName, Location)

      i. Display name of female Patient admitted to hospital at Pokhara.
      ii. Delete the record of Doctors whose salary is greater than average salary of doctors.
      iii. Increase the salary of doctors by 18.5% who are not from Kathmandu.
      iv. Find the average salary of Doctors for each address who have average salary more than 55K.

b) Write the SQL statements for the following queries by reference of Hotal-details relation. 7

hotel_id	hotel_name	est_year	hotel_star	hotel_worth
1	Hyatt	2047	5	1500000
2	Hotel Ktm	2043	3	500000
3	Fulbari	2058	5	2000000
4	Yak and Yeti	2052	4	1100000
5	Hotel Chitwan	2055	3	700000

i. Create a database named hotel and table relation.

ii. Create a view named Price which shows hotel name and its worth.

iii. Modify the data so that Hotel Chitwan is now four star level.

iv. Delete the records of all hotels having worth more than 900000.

3. a) Explain how you would normalize an e-commerce database that tracks customers, orders, and products. 7

b) Explain Data Constraints and its types with examples. 8

4. a) What is multi-valued dependency? Illustrate the advantage of 4NF with suitable example. 8

b) As a database administrator, discuss how Access control and Authentication can be implemented across database systems. 7

5. a) Define query optimization. What are the basic steps of query processing with the help of a diagram. 8

b) What are different recovery scheme? Explain deferred database modification with suitable examples. 7

6. a) Construct a B+-tree for the following set of key values: 7

(8, 10, 12, 14, 115, 16, 20, 24, 28, 34)

Assume that the tree is initially empty and values are added in ascending order. Construct B+ trees for the case where the number of pointers that will fit in one node is **Four**. Also show the form of the tree after deletion of **'24'**.

b) Explain the Two-Phase Locking (2PL) protocol and how it guarantees serializability in concurrent transactions. 8

7. Write short notes on: **(Any two)** 2×5

a) Dense and Sparse index

b) Trigger

c) Distributed Database

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2025
Programme: BE		Full Marks : 100
Course: Database Management System		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) How DBMS address the problem of file system? Explain data abstraction in brief. 7

   b) Draw an E-R diagram for a banking enterprise that keeps the information about employee, customer, loan, account and payment. 8

2. a) Consider the relational database of Figure below, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries: 8

   Student (<u>SID</u>, Name, Department, CGPA)
   Course (<u>CID</u>, CourseName, CreditHours)
   Enrollment (<u>SID, CID</u>, Semester, Grade)

   i. Find the names of students who are enrolled in the course with CID = 'CSE101'
   ii. Get the names of all students who have a CGPA greater than 3.0 and are in the 'CE' department.
   iii. List all students who have enrolled in 'Database Systems' and received grade 'A'
   iv. Get the names of students not enrolled in any course.

   b) Explain the working mechanism of "GROUP BY" clause. Differentiate between WHERE and HAVING clause. 7

3. a) Let us consider the following relation 8

   Sailors (<u>sid,</u> sname, rating, age)
   Boats (<u>bid,</u> bname, color)
   Reserves (<u>sid,bid</u>, day)

   Write a SQL statements for the following

   i. Find the records of sailors who have reserved boat number 20 (bid=20).

   ii. Find the name and rating of sailors who have reserved a black or blue boat.
   iii. Find the age of sailors who have reserved boat number 27 on day 7.
   iv. Update the color of the orange boat to red.
   v. Delete the records of sailors whose rating is less than 7.

   b) Explain Domain, Entity, and Referential Integrity Constraints with suitable SQL examples. 7

4. a) What is normalization? Why it is needed? How can you say that the table is in second and third normal form? Explain with example. 8

   b) Differentiate between authentication and authorization. How encryption and decryption occur in private key and public key cryptography? 7

5. a) Explain the steps involved in Query Processing and Optimization. 7

   b) Create B+ tree of order 4 with following data (4,9,16,25,1,20,13,15,10,11,12). Assume that tree is initially empty and values are added in ascending in order. 8

6. a) What are the different types of failure in database? Differentiate between immediate database modification and deferred database modification. 7

   b) Define serial and non-serial schedule. Explain conflict serializability with example. 8

7. Write short notes on: **(Any two)** 2×5

   a) Data dictionary
   b) Dense and Sparse index
   c) Distributed Databases

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2025
Programme: BE Full Marks : 100
Course: Database Management Systems Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define schemas and instances. What are the major functions of Database Manager? Discuss. 7

   b) Explain the benefits of data model. Construct an ER diagram for keeping records for Library Management System. 8

2. a) Suppose we have the following relation: 8

   Employee(person_name, street, city)

   Works (person_name, company_name, salary)

   Company (company_name, city)

   Write relational algebra expressions for the following queries:

   i. Find names of all employees who live in 'Butwal' and whose salary is less than Rs. 50,000.
   ii. Find the names of all employees who work for "Nabil Bank Limited".
   iii. Find the names and cities of residence of all employees who work for "Global bank".
   iv. Update the salary of all employees by 10% increment.

   b) Define stored procedure. What is the advantage of the stored procedure? Explain how stored procedures are created in SQL. 7

3. a) Consider the relation **Actress_Details** and write the SQL statements for the following queries: 7

Players_id	Actress_name	Debut_year	Recent_release	Actress_fee
1	Renu	2010	Samay	400000
2	Sita	2022	Radha	300000
3	Geeta	2001	Mato	600000
4	Amita	1990	Man	700000
5	Karishma	1989	Prem	100000

   i. Create the table Actress_details relation.

   ii. Delete the data of actress whose recent release is Prem.
   iii. Modify the database so that Renu's new release is "Win the Race" film.
   iv. Insert a new record in the above table.

   b) What are the different types of integrity Constraints? Explain with examples. 8

4. a) What is database normalization? Discuss normalization process with a suitable example until it satisfies 3 NF. 7

   b) What are the needs of security? Explain about the access control, authorization and authentication. 8

5. a) Consider the relation schema in 2(a). Write the relational algebra expression for the query "Find the names of all employees who works in Pokhara". Construct the initial operator tree and final efficient operator tree after applying transformation rules. 8

   b) What is file organization? Explain how you organize files using hash index. 7

6. a) What is Crash Recovery? Explain log based recovery method with example. 7

   b) What do you understand by concurrency control? Explain two phase locking protocol with examples. 8

7. Write short notes on: **(Any two)** 2×5

   a) Data Abstraction
   b) ACID properties
   c) Distributed database